

Reasoning report: Wild type against an isogenic BACH2 knockout: iPSC to T lineage, day 40

Loop example-ipsc-tcell-bach2 · request ipsc-tcell-bach2-d40-001 · 27 decisions · 25 of 30 iterations · ended in protocol_succeeded

What this report is. It records why this loop made each choice, with the references and rationales behind them. The bioreactor was a **synthetic stand-in**, so nothing here is biological evidence, and the decisions were authored by deterministic policy (not a model). Read a recovered parameter as evidence that the search works, not as a finding about any gene or any cell.

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The question and the envelope

Autonomy	autonomous
Reactor	synthetic_standin
Stand-in	ipsc_tcell
Iteration budget	30

Outcome

Final verdict **SUCCESS**: SUCCESS: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT', 'BACH2_KO']; QC PASS.



Final target_cells_per_input_cell per arm, in cells/input_cell, against the required value. Bars are the best configuration each arm ended on.

ARM	OBSERVED	REQUIRED	TARGET	QC
WT	25.744	25	MET	all pass

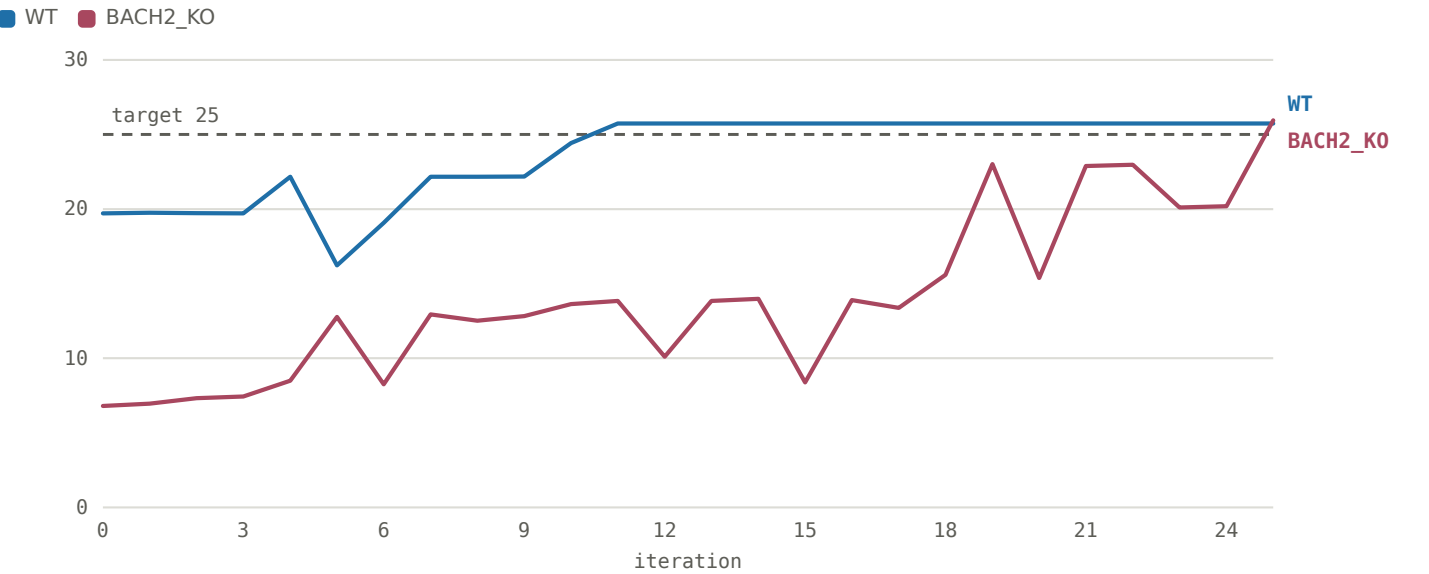
ARM	OBSERVED	REQUIRED	TARGET	QC
BACH2_KO	25.954	25	MET	all pass

Best reading reached per arm

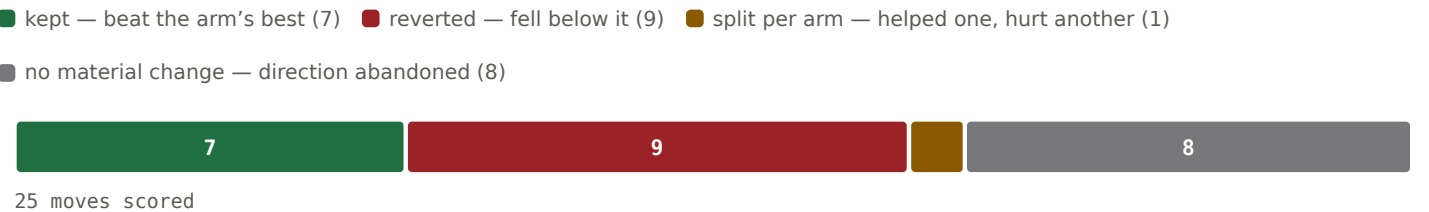
ARM	BEST	ITERATION	PROTOCOL
BACH2_KO	25.954	25	ipsc-tcell-bach2-d40-001-it25
WT	25.744	11	ipsc-tcell-bach2-d40-001-it11

How the search moved

The optimiser is a bounded coordinate search with backtracking. It moves one lever per arm per iteration, scores it against that arm's best reading so far, and reverts anything that does not beat it. It has no model of the process and cannot see interactions between levers.

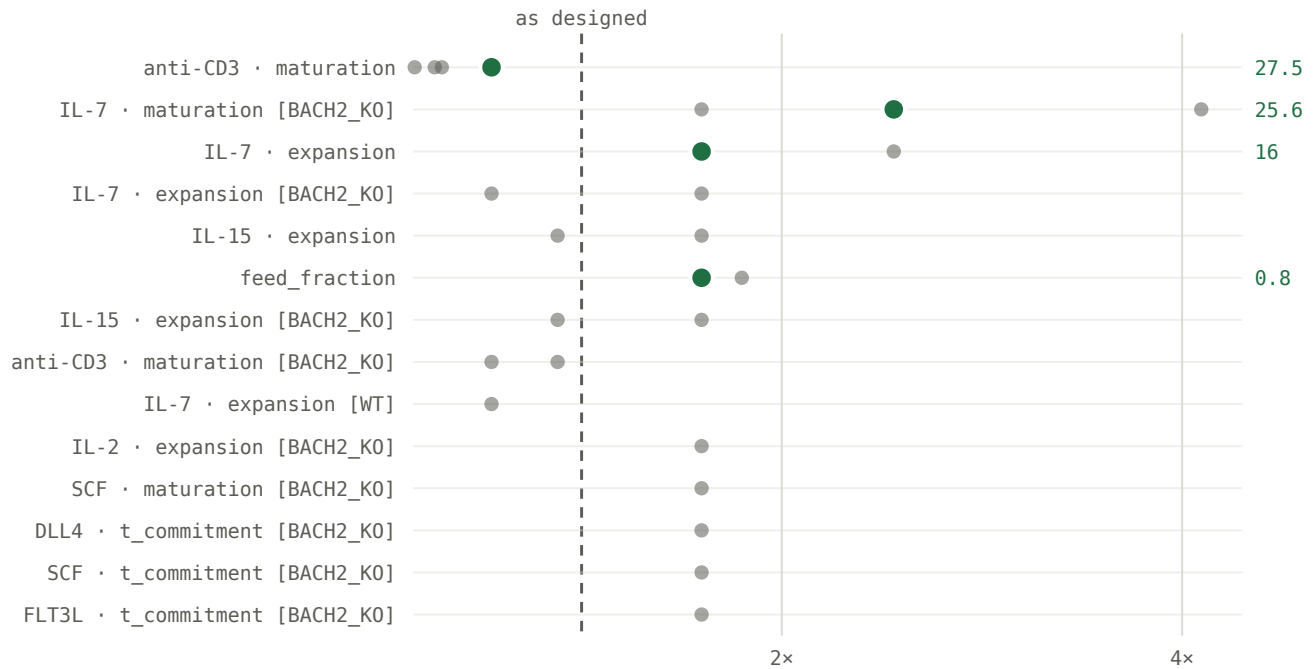


One point per stand-in run. The optimiser moves one lever per arm per iteration and reverts any move that does not beat that arm’s best reading, which is why a line can step back and then recover. The dashed rule is the target the request asked for.



Every move the search made, by what the next run said about it. A reverted move is not a failure of the search — it is the search refusing to keep a change that did not pay, which is what stops it drifting away from a good configuration.

■ a value tried ■ best value found



Each row is one lever, with every value the search tried on it, as a multiple of what the designed protocol started with. A shared absolute axis would be meaningless because the levers carry different units. The filled marker is the value that gave that lever its best reading, printed in its own units on the right.

What the search established

- IL-7 cannot be set to one shared value: moving it 16 -> 25.6 ng/mL improved BACH2_KO and degraded WT. The arms are being given separate values of this parameter from here on.

Reading per iteration

ITERATION	BACH2_KO	WT
0	6.796	19.715
1	6.953	19.758
2	7.326	19.731
3	7.432	19.708
4	8.496	22.174
5	12.764	16.23
6	8.259	19.081
7	12.934	22.174
8	12.518	22.174
9	12.832	22.188
10	13.625	24.43
11	13.85	25.744

	ITERATION	BACH2_KO	WT
	12	10.097	25.744
	13	13.85	25.744
	14	13.986	25.744
	15	8.387	25.744
	16	13.893	25.744
	17	13.371	25.744
	18	15.595	25.744
	19	23.014	25.744
	20	15.383	25.744
	21	22.893	25.744
	22	22.971	25.744
	23	20.104	25.744
	24	20.193	25.744
	25	25.954	25.744

Best reading per arm

ARM	BEST	AT ITERATION
BACH2_KO	25.954	25
WT	25.744	11

Every lever the search touched

PARAMETER	SCOPE	STEP	KIND	VALUES TRIED	EXHAUSTED	BEST
DLL4	BACH2_KO	S12	dose	800	up	
SCF	BACH2_KO	S13	dose	80	up	
FLT3L	BACH2_KO	S14	dose	16	up	
anti-CD3	BACH2_KO	S18	dose	7.321, 11.713	down, up	
anti-CD3	all arms	S18	dose	27.5, 15.125, 8.319, 13.31	down, up	27.5
IL-7	BACH2_KO	S19	dose	16, 25.6, 40.96	up	25.6
SCF	BACH2_KO	S20	dose	80	up	
IL-7	BACH2_KO	S23	dose	40.96, 14.08	down, up	
IL-7	WT	S23	dose	8.8	down	

PARAMETER	SCOPE	STEP	KIND	VALUES TRIED	EXHAUSTED	BEST
IL-7	all arms	S23	dose	16, 25.6	down, up	16
IL-15	BACH2_KO	S24	dose	25.6, 14.08	down, up	
IL-15	all arms	S24	dose	16, 8.8	down, up	
IL-2	BACH2_KO	S25	dose	80	up	
feed_fraction	all arms	culture:feed_fraction	culture	0.8, 0.9	—	0.8

Change log, iteration by iteration

INTO ITERATION	STEP	CHANGE	ADDRESSES
1	S18	anti-CD3: 50 -> 27.5 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it0
2	S18	anti-CD3: 27.5 -> 15.125 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it1
3	S18	anti-CD3: 15.125 -> 8.3188 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it2
4	S23	IL-7: 10 -> 16 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it3
5	S23	IL-7: 16 -> 25.6 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it4
6	S23	IL-7: reverted 25.6 -> 16 ng/mL (all arms) because the result conflict	brief-ipsc-tcell-bach2-d40-001-it5
6	S23	IL-7: 16 -> 8.8 ng/mL (WT)	brief-ipsc-tcell-bach2-d40-001-it5
6	S23	IL-7: 25.6 -> 40.96 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it5
7	S23	IL-7: reverted 8.8 -> 16 ng/mL (WT) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it6
7	S23	IL-7: reverted 40.96 -> 25.6 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it6
7	S24	IL-15: 10 -> 16 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it6
8	S24	IL-15: 16 -> 8.8 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it7
9	S24	IL-15: reverted 8.8 -> 16 ng/mL (all arms) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it8
9	S18	anti-CD3: 8.3188 -> 13.3101 ng/mL (all arms)	brief-ipsc-tcell-bach2-d40-001-it8
10	culture:feed_fraction	feed_fraction: 0.5 -> 0.8 fraction (all arms)	brief-ipsc-tcell-bach2-d40-001-it9

INTO ITERATION	STEP	CHANGE	ADDRESSES
11	culture:feed_fraction	feed_fraction: 0.8 -> 0.9 fraction (all arms)	brief-ipsc-tcell-bach2-d40-001-it10
12	S24	IL-15: 16 -> 25.6 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it11
13	S24	IL-15: reverted 25.6 -> 16 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it12
13	S24	IL-15: 25.6 -> 14.08 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it12
14	S18	anti-CD3: 13.3101 -> 7.3206 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it13
15	S23	IL-7: 25.6 -> 14.08 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it14
16	S23	IL-7: reverted 14.08 -> 25.6 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it15
16	S18	anti-CD3: 7.3206 -> 11.713 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it15
17	S25	IL-2: 50 -> 80 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it16
18	S25	IL-2: reverted 80 -> 50 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it17
18	S19	IL-7: 10 -> 16 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it17
19	S19	IL-7: 16 -> 25.6 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it18
20	S19	IL-7: 25.6 -> 40.96 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it19
21	S19	IL-7: reverted 40.96 -> 25.6 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it20
21	S20	SCF: 50 -> 80 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it20
22	S12	DLL4: 500 -> 800 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it21
23	S13	SCF: 50 -> 80 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it22
24	S13	SCF: reverted 80 -> 50 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it23
24	S14	FLT3L: 10 -> 16 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2-d40-001-it23
25	S14	FLT3L: reverted 16 -> 10 ng/mL (BACH2_KO) because the result made it worse	brief-ipsc-tcell-bach2-d40-001-it24

INTO STEP ITERATION		CHANGE	ADDRESSES
25	S15	IL-7: 10 -> 16 ng/mL (BACH2_KO)	brief-ipsc-tcell-bach2- d40-001-it24

Every decision, with its reasoning

1. request_bioinformatics — decision-00, iteration 0

Authored by

Status

Spends an iteration

Routed to

Validated against

deterministic policy (not a model)

committed

no

bioinformatics

revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. The engine runs one annotation lookup per loop whenever the request declares a gene arm and the envelope permits an information action. The rule exists because an analysis reports a per-arm result but not a cause: a shared change addresses what the arms have in common and leaves the engineered arm unexplained, while annotation costs nothing and cannot spend an iteration. A lookup that returns nothing is also informative, so the rule does not depend on finding an entry.

Stated reason

An annotation lookup for BACH2 runs before the first revision: it is offline, spends no iteration, and may scope the change to one arm.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
revise_protocol	Allowed here, but the rule gathers first: revising now would spend an iteration on a shared change while the engineered arm stayed unexplained.

Hypotheses raised, each with its basis

• **OH1** The engineered arm needs different conditions from the control rather than simply growing more slowly under the same ones.

Basis:

◦ WT observed 19.7151 vs required 25.0 under an identical schedule

◦ BACH2_KO observed 6.79625 vs required 25.0 under an identical schedule

Would be tested by: A per-arm adjustment that changes only the implicated parameters

Tool results cited

TOOL	PURPOSE	RESULT
bioinformatics.annotate	perturbation effects and implicated levers	4 levers implicated: expansion duration (shorten, inference), TCR/activation signal strength (decrease, inference), IL-7 dose (increase, inference), IL-15 dose (increase, inference)

2. revise_protocol — decision-01, iteration 1

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC FAIL. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 1 protocol ipsc-tcell-bach2-d40-001-it0 did not work: target:WT NOT_MET (observed 19.7151, required 25.0); target:BACH2_KO NOT_MET (observed 6.79625, required 25.0); qc-residual-pluripotency FAIL (observed 0.604, required <= 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Exit from pluripotency is incomplete: undifferentiated cells persist to harvest.
Basis:
◦ residual pluripotency 0.604% vs limit 0.5%
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
◦ target_cells_per_input_cell 19.7151 vs required 25.0 (shortfall 21%) with purity in specification
◦ target_cells_per_input_cell 6.79625 vs required 25.0 (shortfall 73%) with purity in specification
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H03** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
◦ target_cells_per_input_cell: BACH2_KO 6.79625 vs WT 19.7151 (ratio 0.3447; single replicate; change must exceed 10% of the control)
Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it0

Iteration 1 protocol ipsc-tcell-bach2-d40-001-it0 did not work: target:WT NOT_MET (observed 19.7151, required 25.0); target:BACH2_KO NOT_MET (observed 6.79625, required 25.0); qc-residual-pluripotency FAIL (observed 0.604, required ≤ 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 19.715, required 25.0
- target:BACH2_KO **NOT_MET** — observed 6.796, required 25.0
- qc-residual-pluripotency **FAIL** — observed 0.604, required ≤ 0.5

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
expansion -> induction transition day	revisit		residual pluripotent cells persist to harvest
VEGF dose/window	revisit	S07	exit from pluripotency incomplete
FGF2 dose/window	revisit	S08	exit from pluripotency incomplete
SCF dose/window	revisit	S09	exit from pluripotency incomplete
SB431542 dose/window	revisit	S10	exit from pluripotency incomplete
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement

3. revise_protocol — decision-02, iteration 2

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC FAIL. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 2 protocol ipsc-tcell-bach2-d40-001-it1 did not work: target:WT NOT_MET (observed 19.7577, required 25.0); target:BACH2_KO NOT_MET (observed 6.95291, required 25.0); qc-residual-pluripotency FAIL (observed 0.59, required <= 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Exit from pluripotency is incomplete: undifferentiated cells persist to harvest.
Basis:
◦ residual pluripotency 0.59% vs limit 0.5%
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
◦ target_cells_per_input_cell 19.7577 vs required 25.0 (shortfall 21%) with purity in specification
◦ target_cells_per_input_cell 6.95291 vs required 25.0 (shortfall 72%) with purity in specification
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H03** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
◦ target_cells_per_input_cell: BACH2_KO 6.95291 vs WT 19.7577 (ratio 0.3519; single replicate; change must exceed 10% of the control)
Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it1

Iteration 2 protocol ipsc-tcell-bach2-d40-001-it1 did not work: target:WT NOT_MET (observed 19.7577, required 25.0); target:BACH2_KO NOT_MET (observed 6.95291, required 25.0); qc-residual-pluripotency FAIL (observed 0.59, required ≤ 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 19.758, required 25.0
- target:BACH2_KO **NOT_MET** — observed 6.953, required 25.0
- qc-residual-pluripotency **FAIL** — observed 0.59, required ≤ 0.5

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
expansion -> induction transition day	revisit		residual pluripotent cells persist to harvest
VEGF dose/window	revisit	S07	exit from pluripotency incomplete
FGF2 dose/window	revisit	S08	exit from pluripotency incomplete
SCF dose/window	revisit	S09	exit from pluripotency incomplete
SB431542 dose/window	revisit	S10	exit from pluripotency incomplete
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement

4. revise_protocol — decision-03, iteration 3

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC FAIL. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 3 protocol ipsc-tcell-bach2-d40-001-it2 did not work: target:WT NOT_MET (observed 19.7307, required 25.0); target:BACH2_KO NOT_MET (observed 7.32619, required 25.0); qc-residual-pluripotency FAIL (observed 0.56, required <= 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Exit from pluripotency is incomplete: undifferentiated cells persist to harvest.
Basis:
◦ residual pluripotency 0.56% vs limit 0.5%
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
◦ target_cells_per_input_cell 19.7307 vs required 25.0 (shortfall 21%) with purity in specification
◦ target_cells_per_input_cell 7.32619 vs required 25.0 (shortfall 71%) with purity in specification
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H03** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
◦ target_cells_per_input_cell: BACH2_KO 7.32619 vs WT 19.7307 (ratio 0.3713; single replicate; change must exceed 10% of the control)
Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it2

Iteration 3 protocol ipsc-tcell-bach2-d40-001-it2 did not work: target:WT NOT_MET (observed 19.7307, required 25.0); target:BACH2_KO NOT_MET (observed 7.32619, required 25.0); qc-residual-pluripotency FAIL (observed 0.56, required ≤ 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 19.731, required 25.0
- target:BACH2_KO **NOT_MET** — observed 7.326, required 25.0
- qc-residual-pluripotency **FAIL** — observed 0.56, required ≤ 0.5

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
expansion -> induction transition day	revisit		residual pluripotent cells persist to harvest
VEGF dose/window	revisit	S07	exit from pluripotency incomplete
FGF2 dose/window	revisit	S08	exit from pluripotency incomplete
SCF dose/window	revisit	S09	exit from pluripotency incomplete
SB431542 dose/window	revisit	S10	exit from pluripotency incomplete
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement

5. revise_protocol — decision-04, iteration 4

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC FAIL. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 4 protocol ipsc-tcell-bach2-d40-001-it3 did not work: target:WT NOT_MET (observed 19.7083, required 25.0); target:BACH2_KO NOT_MET (observed 7.4321, required 25.0); qc-residual-pluripotency FAIL (observed 0.552, required <= 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Exit from pluripotency is incomplete: undifferentiated cells persist to harvest.
Basis:
◦ residual pluripotency 0.552% vs limit 0.5%
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
◦ target_cells_per_input_cell 19.7083 vs required 25.0 (shortfall 21%) with purity in specification
◦ target_cells_per_input_cell 7.4321 vs required 25.0 (shortfall 70%) with purity in specification
Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H03** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
◦ target_cells_per_input_cell: BACH2_KO 7.4321 vs WT 19.7083 (ratio 0.3771; single replicate; change must exceed 10% of the control)
Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it3

Iteration 4 protocol ipsc-tcell-bach2-d40-001-it3 did not work: target:WT NOT_MET (observed 19.7083, required 25.0); target:BACH2_KO NOT_MET (observed 7.4321, required 25.0); qc-residual-pluripotency FAIL (observed 0.552, required ≤ 0.5). Most likely reasons: H01 [residual_pluripotency, moderate] Exit from pluripotency is incomplete: undifferentiated cells persist to harvest. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H03 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 19.708, required 25.0
- target:BACH2_KO **NOT_MET** — observed 7.432, required 25.0
- qc-residual-pluripotency **FAIL** — observed 0.552, required ≤ 0.5

Must stay fixed

- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
expansion -> induction transition day	revisit		residual pluripotent cells persist to harvest
VEGF dose/window	revisit	S07	exit from pluripotency incomplete
FGF2 dose/window	revisit	S08	exit from pluripotency incomplete
SCF dose/window	revisit	S09	exit from pluripotency incomplete
SB431542 dose/window	revisit	S10	exit from pluripotency incomplete
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement

6. revise_protocol — decision-05, iteration 5

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 5 protocol ipsc-tcell-bach2-d40-001-it4 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 8.49579, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.1736 vs required 25.0 (shortfall 11%) with purity in specification
 - target_cells_per_input_cell 8.49579 vs required 25.0 (shortfall 66%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 8.49579 vs WT 22.1736 (ratio 0.3831; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it4

Iteration 5 protocol ipsc-tcell-bach2-d40-001-it4 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 8.49579, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 22.174, required 25.0
- target:BACH2_KO **NOT_MET** — observed 8.496, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement
SB431542 dose/window for BACH2_KO	revisit	S10	genotype-dependent requirement
DLL4 dose/window for BACH2_KO	revisit	S12	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S13	genotype-dependent requirement
FLT3L dose/window for BACH2_KO	revisit	S14	genotype-dependent requirement
IL-7 dose/window for BACH2_KO	revisit	S15	genotype-dependent requirement

7. revise_protocol — decision-06, iteration 6

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 6 protocol ipsc-tcell-bach2-d40-001-it5 did not work: target:WT NOT_MET (observed 16.2297, required 25.0); target:BACH2_KO NOT_MET (observed 12.7636, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 16.2297 vs required 25.0 (shortfall 35%) with purity in specification
 - target_cells_per_input_cell 12.7636 vs required 25.0 (shortfall 49%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 12.7636 vs WT 16.2297 (ratio 0.7864; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it5

Iteration 6 protocol ipsc-tcell-bach2-d40-001-it5 did not work: target:WT NOT_MET (observed 16.2297, required 25.0); target:BACH2_KO NOT_MET (observed 12.7636, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 16.23, required 25.0
- target:BACH2_KO **NOT_MET** — observed 12.764, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
VEGF dose/window for BACH2_KO	revisit	S07	genotype-dependent requirement
FGF2 dose/window for BACH2_KO	revisit	S08	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S09	genotype-dependent requirement
SB431542 dose/window for BACH2_KO	revisit	S10	genotype-dependent requirement
DLL4 dose/window for BACH2_KO	revisit	S12	genotype-dependent requirement
SCF dose/window for BACH2_KO	revisit	S13	genotype-dependent requirement
FLT3L dose/window for BACH2_KO	revisit	S14	genotype-dependent requirement
IL-7 dose/window for BACH2_KO	revisit	S15	genotype-dependent requirement

8. revise_protocol — decision-07, iteration 7

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 7 protocol ipsc-tcell-bach2-d40-001-it6 did not work: target:WT NOT_MET (observed 19.0806, required 25.0); target:BACH2_KO NOT_MET (observed 8.25915, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 19.0806 vs required 25.0 (shortfall 24%) with purity in specification
 - target_cells_per_input_cell 8.25915 vs required 25.0 (shortfall 67%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 8.25915 vs WT 19.0806 (ratio 0.4329; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it6

Iteration 7 protocol ipsc-tcell-bach2-d40-001-it6 did not work: target:WT NOT_MET (observed 19.0806, required 25.0); target:BACH2_KO NOT_MET (observed 8.25915, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 19.081, required 25.0
- target:BACH2_KO **NOT_MET** — observed 8.259, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
BACH2_KO adjustment adj-it6-02	revisit	S23	BACH2_KO only. Search step from the search itself, after scoring the previous move. Lever: 'IL-7 for BACH2_KO' increase. Basis given for the lever: Split from a shared value after it improved BACH2_KO and degraded WT. Each arm now moves in the direction its own result asked for.. Moved from 25.6 to 40.96 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it5. This value came from a bounded search step, not from any publication, so it is a design choice.

9. revise_protocol — decision-08, iteration 8

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 8 protocol ipsc-tcell-bach2-d40-001-it7 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 12.9341, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.1736 vs required 25.0 (shortfall 11%) with purity in specification
 - target_cells_per_input_cell 12.9341 vs required 25.0 (shortfall 48%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 12.9341 vs WT 22.1736 (ratio 0.5833; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it7

Iteration 8 protocol ipsc-tcell-bach2-d40-001-it7 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 12.9341, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 22.174, required 25.0
- target:BACH2_KO **NOT_MET** — observed 12.934, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.

10. revise_protocol — decision-09, iteration 9

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 9 protocol ipsc-tcell-bach2-d40-001-it8 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 12.5184, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.1736 vs required 25.0 (shortfall 11%) with purity in specification
 - target_cells_per_input_cell 12.5184 vs required 25.0 (shortfall 50%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 12.5184 vs WT 22.1736 (ratio 0.5646; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it8

Iteration 9 protocol ipsc-tcell-bach2-d40-001-it8 did not work: target:WT NOT_MET (observed 22.1736, required 25.0); target:BACH2_KO NOT_MET (observed 12.5184, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 22.174, required 25.0
- target:BACH2_KO **NOT_MET** — observed 12.518, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.

11. revise_protocol — decision-10, iteration 10

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 10 protocol ipsc-tcell-bach2-d40-001-it9 did not work: target:WT NOT_MET (observed 22.1882, required 25.0); target:BACH2_KO NOT_MET (observed 12.8324, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.1882 vs required 25.0 (shortfall 11%) with purity in specification
 - target_cells_per_input_cell 12.8324 vs required 25.0 (shortfall 49%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 12.8324 vs WT 22.1882 (ratio 0.5783; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it9

Iteration 10 protocol ipsc-tcell-bach2-d40-001-it9 did not work: target:WT NOT_MET (observed 22.1882, required 25.0); target:BACH2_KO NOT_MET (observed 12.8324, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 22.188, required 25.0
- target:BACH2_KO **NOT_MET** — observed 12.832, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.

12. revise_protocol — decision-11, iteration 11

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by no arm; not met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 11 protocol ipsc-tcell-bach2-d40-001-it10 did not work: target:WT NOT_MET (observed 24.4302, required 25.0); target:BACH2_KO NOT_MET (observed 13.625, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 24.4302 vs required 25.0 (shortfall 2%) with purity in specification
 - target_cells_per_input_cell 13.625 vs required 25.0 (shortfall 46%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 13.625 vs WT 24.4302 (ratio 0.5577; single replicate; change must exceed 10% of the control)
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it10

Iteration 11 protocol ipsc-tcell-bach2-d40-001-it10 did not work: target:WT NOT_MET (observed 24.4302, required 25.0); target:BACH2_KO NOT_MET (observed 13.625, required 25.0). Most likely reasons: H01 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are. H02 [genotype_specific, low] BACH2_KO (BACH2 knockout) produces less than WT under the same protocol: beyond the shared shortfall, the edit changes what this arm needs.

Failed criteria

- target:WT **NOT_MET** — observed 24.43, required 25.0
- target:BACH2_KO **NOT_MET** — observed 13.625, required 25.0

Must stay fixed

- qc-viability passed in every arm; do not trade it away

- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away
- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.

13. revise_protocol — decision-12, iteration 12

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 12 protocol ipsc-tcell-bach2-d40-001-it11 did not work: target:BACH2_KO NOT_MET (observed 13.8498, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 13.8498 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 13.8498 vs required 25.0 (shortfall 45%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it11

Iteration 12 protocol ipsc-tcell-bach2-d40-001-it11 did not work: target:BACH2_KO NOT_MET (observed 13.8498, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 13.85, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

14. revise_protocol — decision-13, iteration 13

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 13 protocol ipsc-tcell-bach2-d40-001-it12 did not work: target:BACH2_KO NOT_MET (observed 10.0969, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 10.0969 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 10.0969 vs required 25.0 (shortfall 60%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it12

Iteration 13 protocol ipsc-tcell-bach2-d40-001-it12 did not work: target:BACH2_KO NOT_MET (observed 10.0969, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 10.097, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it12-02	revisit	S24	BACH2_KO only. Search step from the bioinformatics annotation (annotation confidence: inference). Lever: 'IL-15 dose' increase. Basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 16 to 25.6 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it11. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

15. revise_protocol — decision-14, iteration 14

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 14 protocol ipsc-tcell-bach2-d40-001-it13 did not work: target:BACH2_KO NOT_MET (observed 13.8498, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 13.8498 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 13.8498 vs required 25.0 (shortfall 45%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it13

Iteration 14 protocol ipsc-tcell-bach2-d40-001-it13 did not work: target:BACH2_KO NOT_MET (observed 13.8498, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 13.85, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

16. revise_protocol — decision-15, iteration 15

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 15 protocol ipsc-tcell-bach2-d40-001-it14 did not work: target:BACH2_KO NOT_MET (observed 13.986, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 13.986 vs WT 25.7444
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 13.986 vs required 25.0 (shortfall 44%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it14

Iteration 15 protocol ipsc-tcell-bach2-d40-001-it14 did not work: target:BACH2_KO NOT_MET (observed 13.986, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 13.986, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it7-03	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it14-03	revisit	S18	BACH2_KO only. Search step from the bioinformatics annotation (annotation confidence: inference). Lever: 'TCR/activation signal strength' decrease. Basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 13.3101 to 7.3206 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it13. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

17. revise_protocol — decision-16, iteration 16

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 16 protocol ipsc-tcell-bach2-d40-001-it15 did not work: target:BACH2_KO NOT_MET (observed 8.38666, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 8.38666 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 8.38666 vs required 25.0 (shortfall 66%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it15

Iteration 16 protocol ipsc-tcell-bach2-d40-001-it15 did not work: target:BACH2_KO NOT_MET (observed 8.38666, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 8.387, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj- it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj- it14-03	revisit	S18	BACH2_KO only. Search step from the bioinformatics annotation (annotation confidence: inference). Lever: 'TCR/activation signal strength' decrease. Basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 13.3101 to 7.3206 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it13. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj- it15-04	revisit	S23	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-7 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from tsukumo2013 and roychoudhuri2016, both of which tie BACH2 to naive/memory maintenance: homeostatic cytokine support is the usual protocol lever for that state. No cited paper reports an IL-7 dose for a BACH2 knockout.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it14. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage

PARAMETER	DIRECTION	STEP	BASIS
IL-2 dose/window	revisit	S25	drives output in the production stage

18. revise_protocol — decision-17, iteration 17

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 17 protocol ipsc-tcell-bach2-d40-001-it16 did not work: target:BACH2_KO NOT_MET (observed 13.8928, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 13.8928 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 13.8928 vs required 25.0 (shortfall 44%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it16

Iteration 17 protocol ipsc-tcell-bach2-d40-001-it16 did not work: target:BACH2_KO NOT_MET (observed 13.8928, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 13.893, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

19. revise_protocol — decision-18, iteration 18

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 18 protocol ipsc-tcell-bach2-d40-001-it17 did not work: target:BACH2_KO NOT_MET (observed 13.3706, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 13.3706 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 13.3706 vs required 25.0 (shortfall 47%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it17

Iteration 18 protocol ipsc-tcell-bach2-d40-001-it17 did not work: target:BACH2_KO NOT_MET (observed 13.3706, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 13.371, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it17-04	revisit	S25	BACH2_KO only. Search step from unimplicated_search. Lever: 'IL-2 (expansion)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it16. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage

PARAMETER	DIRECTION	STEP	BASIS
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

20. revise_protocol — decision-19, iteration 19

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 19 protocol ipsc-tcell-bach2-d40-001-it18 did not work: target:BACH2_KO NOT_MET (observed 15.5951, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 15.5951 vs WT 25.7444
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 15.5951 vs required 25.0 (shortfall 38%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it18

Iteration 19 protocol ipsc-tcell-bach2-d40-001-it18 did not work: target:BACH2_KO NOT_MET (observed 15.5951, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 15.595, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it18-05	revisit	S19	BACH2_KO only. Search step from unimplicated_search. Lever: 'IL-7 (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 10 to 16 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it17. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage

PARAMETER	DIRECTION	STEP	BASIS
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

21. revise_protocol — decision-20, iteration 20

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 20 protocol ipsc-tcell-bach2-d40-001-it19 did not work: target:BACH2_KO NOT_MET (observed 23.0142, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 23.0142 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 23.0142 vs required 25.0 (shortfall 8%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it19

Iteration 20 protocol ipsc-tcell-bach2-d40-001-it19 did not work: target:BACH2_KO NOT_MET (observed 23.0142, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 23.014, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it19-05	revisit	S19	BACH2_KO only. Search step from the search itself, after scoring the previous move. Lever: 'IL-7 (maturation)' increase. Basis given for the lever: Continuing an ascent: the previous step on this lever (10 -> 16 ng/mL) improved BACH2_KO, so the same direction is tried again. Original basis: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 16 to 25.6 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it18. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation

PARAMETER	DIRECTION	STEP	BASIS
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

22. revise_protocol — decision-21, iteration 21

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 21 protocol ipsc-tcell-bach2-d40-001-it20 did not work: target:BACH2_KO NOT_MET (observed 15.3832, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 15.3832 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 15.3832 vs required 25.0 (shortfall 38%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it20

Iteration 21 protocol ipsc-tcell-bach2-d40-001-it20 did not work: target:BACH2_KO NOT_MET (observed 15.3832, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 15.383, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it20-05	revisit	S19	BACH2_KO only. Search step from the search itself, after scoring the previous move. Lever: 'IL-7 (maturation)' increase. Basis given for the lever: Continuing an ascent: the previous step on this lever (16 -> 25.6 ng/mL) improved BACH2_KO, so the same direction is tried again. Original basis: Continuing an ascent: the previous step on this lever (10 -> 16 ng/mL) improved BACH2_KO, so the same direction is tried again. Original basis: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 25.6 to 40.96 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it19. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity

PARAMETER	DIRECTION	STEP	BASIS
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

23. revise_protocol — decision-22, iteration 22

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 22 protocol ipsc-tcell-bach2-d40-001-it21 did not work: target:BACH2_KO NOT_MET (observed 22.8929, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 22.8929 vs WT 25.7444
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.8929 vs required 25.0 (shortfall 8%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it21

Iteration 22 protocol ipsc-tcell-bach2-d40-001-it21 did not work: target:BACH2_KO NOT_MET (observed 22.8929, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 22.893, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it21-05	revisit	S19	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it21-06	revisit	S20	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation

PARAMETER	DIRECTION	STEP	BASIS
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

24. revise_protocol — decision-23, iteration 23

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 23 protocol ipsc-tcell-bach2-d40-001-it22 did not work: target:BACH2_KO NOT_MET (observed 22.9713, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.
Basis:
 - target_cells_per_input_cell: BACH2_KO 22.9713 vs WT 25.7444
 Would be tested by: A protocol revision that moves the levers this hypothesis names
- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.
Basis:
 - target_cells_per_input_cell 22.9713 vs required 25.0 (shortfall 8%) with purity in specification
 Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it22

Iteration 23 protocol ipsc-tcell-bach2-d40-001-it22 did not work: target:BACH2_KO NOT_MET (observed 22.9713, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 22.971, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it21-05	revisit	S19	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it21-06	revisit	S20	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it22-06	revisit	S12	BACH2_KO only. Search step from unimplicated_search. Lever: 'DLL4 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 500 to 800 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it21. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

25. revise_protocol — decision-24, iteration 24

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 24 protocol ipsc-tcell-bach2-d40-001-it23 did not work: target:BACH2_KO NOT_MET (observed 20.1044, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 20.1044 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 20.1044 vs required 25.0 (shortfall 20%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it23

Iteration 24 protocol ipsc-tcell-bach2-d40-001-it23 did not work: target:BACH2_KO NOT_MET (observed 20.1044, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 20.104, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it21-05	revisit	S19	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj-it21-06	revisit	S20	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it22-06	revisit	S12	BACH2_KO only. Search step from unimplicated_search. Lever: 'DLL4 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 500 to 800 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it21. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it23-07	revisit	S13	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it22. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

26. revise_protocol — decision-25, iteration 25

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	yes
Routed to	literature
Validated against	revise_protocol, request_bioinformatics, request_literature, consult_human, escalate_to_human, repeat_measurement

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict FAILED: FAILED: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT']; not met by ['BACH2_KO']; QC PASS. The envelope allowed ['consult_human', 'escalate_to_human', 'repeat_measurement', 'request_bioinformatics', 'request_literature', 'revise_protocol'] and the policy rule for this verdict selects 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Stated reason

Iteration 25 protocol ipsc-tcell-bach2-d40-001-it24 did not work: target:BACH2_KO NOT_MET (observed 20.1928, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
consult_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
escalate_to_human	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
protocol_succeeded	Refused by the envelope: A failed target or criterion cannot be reported as success.

ACTION	WHY IT WAS NOT CHOSEN
repeat_measurement	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_bioinformatics	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.
request_literature	Allowed for this verdict but not selected: the policy rule for FAILED prefers 'revise_protocol'. At least one arm missed the target or a QC criterion, the measurements are trustworthy, and iteration budget remains. The failure is therefore attributable to the recipe, which is the one thing in the loop that can be changed, so the protocol is revised against a brief naming what failed, which levers the diagnosis implicates and what must stay fixed.

Hypotheses raised, each with its basis

- **H01** The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs.

Basis:

- target_cells_per_input_cell: BACH2_KO 20.1928 vs WT 25.7444

Would be tested by: A protocol revision that moves the levers this hypothesis names

- **H02** Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Basis:

- target_cells_per_input_cell 20.1928 vs required 25.0 (shortfall 19%) with purity in specification

Would be tested by: A protocol revision that moves the levers this hypothesis names

Revision brief brief-ipsc-tcell-bach2-d40-001-it24

Iteration 25 protocol ipsc-tcell-bach2-d40-001-it24 did not work: target:BACH2_KO NOT_MET (observed 20.1928, required 25.0). Most likely reasons: H01 [genotype_specific, moderate] The protocol works for WT but not for BACH2_KO (BACH2 knockout): the edit changes what this arm needs. H02 [expansion, low] Output is quantity-limited: too few cells reach harvest, independent of how pure they are.

Failed criteria

- target:BACH2_KO **NOT_MET** — observed 20.193, required 25.0

Must stay fixed

- WT met the target (target_cells_per_input_cell 25.7444 $\geq$ 25.0); keep its base schedule unless a change is shared on purpose
- qc-viability passed in every arm; do not trade it away
- qc-identity-purity passed in every arm; do not trade it away
- qc-residual-pluripotency passed in every arm; do not trade it away

- qc-sterility passed in every arm; do not trade it away
- qc-mycoplasma passed in every arm; do not trade it away
- qc-endotoxin passed in every arm; do not trade it away
- qc-karyotype passed in every arm; do not trade it away

Levers offered (first 14)

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj- it13-03	revisit	S24	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj- it16-04	revisit	S23	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj- it16-05	revisit	S18	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj- it21-05	revisit	S19	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO adjustment adj- it21-06	revisit	S20	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.

PARAMETER	DIRECTION	STEP	BASIS
BACH2_KO adjustment adj-it22-06	revisit	S12	BACH2_KO only. Search step from unimplicated_search. Lever: 'DLL4 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 500 to 800 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it21. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO adjustment adj-it24-08	revisit	S14	BACH2_KO only. Search step from unimplicated_search. Lever: 'FLT3L (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 10 to 16 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it23. This value came from a bounded search step, not from any publication, so it is a design choice.
seeding density	revisit	S01	more input or better expansion raises output at fixed purity
expansion duration	extend		more cells entering differentiation
production/harvest stage duration	revisit		output accumulates during this stage
IL-7 dose/window	revisit	S23	drives output in the production stage
IL-15 dose/window	revisit	S24	drives output in the production stage
IL-2 dose/window	revisit	S25	drives output in the production stage

27. protocol_succeeded — decision-26, iteration 26

Authored by	deterministic policy (not a model)
Status	committed
Spends an iteration	no
Routed to	human
Validated against	protocol_succeeded, escalate_to_human

Reasoning as written by the deterministic policy, which is a description of a rule

Deterministic policy, not a model. Verdict SUCCESS: SUCCESS: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT', 'BACH2_KO']; QC PASS. The envelope allowed ['escalate_to_human', 'protocol_succeeded'] and the policy rule for this verdict selects 'protocol_succeeded'. Every arm that the request holds to the target met it, and every QC criterion passed. There is nothing further to gather evidence for and nothing left to revise, so the loop reports success. This is not a released process: confirmation runs and human QA sign-off are outside this loop.

Stated reason

All target arms met the desired output and every QC criterion passed. SUCCESS: target target_cells_per_input_cell >= 25 cells/input_cell at day 40 met by ['WT', 'BACH2_KO']; QC PASS. Release still requires human QA sign-off and confirmation runs.

Alternatives considered, and why not

ACTION	WHY IT WAS NOT CHOSEN
escalate_to_human	Allowed for this verdict but not selected: the policy rule for SUCCESS prefers 'protocol_succeeded'. Every arm that the request holds to the target met it, and every QC criterion passed. There is nothing further to gather evidence for and nothing left to revise, so the loop reports success. This is not a released process: confirmation runs and human QA sign-off are outside this loop.
stop_budget	Refused by the envelope: The target was met; stopping on budget would misreport the outcome.

The protocol it ended on

This is the configuration the search held at the end, which under its own accept/reject rule is the best it found. Read the provenance column before reading any number: a design choice is a value the loop chose, not a value anyone measured.

■ Design choice (36)



36 quantities

Every quantity in the final protocol by where its value came from. A bar that is entirely design choice means nothing in the recipe is attributable to a publication, which is a statement about the evidence available, not about the recipe being wrong.

Protocol	ipsc-tcell-bach2-d40-001-it25
Title	Designed iPSC -> T cell schedule to day 40 (WT, BACH2_KO) (iteration 25: IL-7 increase in BACH2_KO; 1 reverted)
Day 0 is	iPSC seeding into the vessel
Approved by	automated stand-in run (no human approved this; no wet-lab run is authorised by this protocol)
Quantities	Design choice 36

Where the arms differ

These are the per-arm deviations the search established. A parameter listed here could not be satisfied by one shared value.

ARM	STEP	VALUE	UNIT	WHY
BACH2_KO	S24	14.08	ng/mL	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO	S23	25.6	ng/mL	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO	S18	11.713	ng/mL	BACH2_KO only. Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO	S19	25.6	ng/mL	BACH2_KO only. Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
BACH2_KO	S20	80	ng/mL	BACH2_KO only. Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.

ARM	STEP	VALUE	UNIT	WHY
BACH2_KO	S12	800	ng/mL	BACH2_KO only. Search step from unimplicated_search. Lever: 'DLL4 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 500 to 800 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it21. This value came from a bounded search step, not from any publication, so it is a design choice.
BACH2_KO	S15	16	ng/mL	BACH2_KO only. Search step from unimplicated_search. Lever: 'IL-7 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 10 to 16 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it24. This value came from a bounded search step, not from any publication, so it is a design choice.

Every quantity, with its provenance

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
culture_system.working_volume	100	mL	Design choice	–	Working volume chosen here for a bench-scale run. Not a retrieved value.
culture_system.parameters.dissolved_oxygen	21	%	Design choice	–	Atmospheric oxygen, chosen here; a hypoxic early phase is a plausible alternative this schedule does not test.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
culture_system.parameters.feed_fraction	0.9	fraction	Design choice	–	Search step from the search itself, after scoring the previous move. Lever: 'feeding regime (feed_fraction)' increase. Basis given for the lever: Continuing an ascent: the previous step on this lever (0.5 -> 0.8 fraction) improved BACH2_KO, WT, so the same direction is tried again. Original basis: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. Explored because the implicated levers are exhausted. A fed-batch split changes how much medium the culture gets and therefore the total yield, so a gain here says something about the vessel, not about the biology.. Moved from 0.8 to 0.9 fraction for every arm. Brief: brief-ipsc-tcell-bach2-d40-001-it10. This value came from a bounded search step, not from any publication, so it is a design choice.
culture_system.parameters.temperature	37	degC	Design choice	–	Standard culture temperature.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
mesoderm.medium	serum-free, xeno-free basal medium with mesoderm supplements	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
mesoderm.S01 (seed)	200,000	cells/mL	Design choice	–	Starting iPSC density chosen here as a mid-range aggregate seeding density. No retrieved value supports it.
mesoderm.S02 (BMP4)	20	ng/mL	Design choice	–	Standard mesoderm-inducing signal in pluripotent-stem-cell haematopoietic differentiation. Dose is a mid-range starting point chosen here, not a retrieved value.
mesoderm.S03 (VEGF)	50	ng/mL	Design choice	L02	Supports endothelial specification and the endothelial-to-haematopoietic transition. Dose chosen here.
mesoderm.S04 (FGF2)	20	ng/mL	Design choice	–	General proliferative support across the early stages. Dose chosen here.
mesoderm.S05 (CHIR99021)	4	uM	Design choice	–	GSK3 inhibitor used as a Wnt agonist for mesoderm patterning, applied as a short pulse. Dose and pulse length chosen here.
hemogenic_endothelium.medium	serum-free basal medium with haematopoietic supplements	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
hemogenic_endothelium.S07 (VEGF)	50	ng/mL	Design choice	L02	Supports endothelial specification and the endothelial-to-haematopoietic transition. Dose chosen here.
hemogenic_endothelium.S08 (FGF2)	20	ng/mL	Design choice	–	General proliferative support across the early stages. Dose chosen here.
hemogenic_endothelium.S09 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.
hemogenic_endothelium.S10 (SB431542)	5	uM	Design choice	–	TGF-beta receptor inhibitor, applied briefly to favour the haematopoietic transition. Dose and window chosen here.
t_commitment.medium	serum-free lymphoid medium on a DLL4-presenting surface	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
t_commitment.S12 (DLL4)	500	ng/mL	Design choice	L01	Notch ligand presentation is the lever that commits progenitors to the T lineage; the cited work reports that providing DLL4 increases T-competent progenitor output. The direction is cited; the surface density here is a design choice.
t_commitment.S13 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
t_commitment.S14 (FLT3L)	10	ng/mL	Design choice	–	Supports lymphoid progenitor expansion alongside SCF. Dose chosen here.
t_commitment.S15 (IL-7)	10	ng/mL	Design choice	L04	The central homeostatic cytokine for T-lineage survival and proliferation, and the lever the cited BACH2 work implicates for maintaining a less differentiated state. Dose chosen here.
t_commitment.S16 (TPO)	20	ng/mL	Design choice	–	Supports early haematopoietic progenitors; withdrawn mid-stage. Dose and window chosen here.
maturation.medium	serum-free lymphoid medium	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
maturation.S18 (anti-CD3)	13.31	ng/mL	Design choice	–	Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for all arms, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 8.3188 to 13.3101 ng/mL for every arm. Brief: brief-ipsc-tcell-bach2-d40-001-it8. This value came from a bounded search step, not from any publication, so it is a design choice.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
maturation.S19 (IL-7)	10	ng/mL	Design choice	L04	The central homeostatic cytokine for T-lineage survival and proliferation, and the lever the cited BACH2 work implicates for maintaining a less differentiated state. Dose chosen here.
maturation.S20 (SCF)	50	ng/mL	Design choice	–	Haematopoietic progenitor survival and proliferation factor. Dose chosen here.
expansion.medium	serum-free lymphoid medium	text	Design choice	–	Medium class chosen here to match the stage goal. No retrieved medium formulation supports it.
expansion.S23 (IL-7)	16	ng/mL	Design choice	–	Reverted to 16 ng/mL: the previous revision moved this to 25.6 and the result conflict. The arms are given separate values of this parameter instead.
expansion.S24 (IL-15)	16	ng/mL	Design choice	–	Reverted to 16 ng/mL: the previous revision moved this to 8.8 and the result made it worse. That direction is now treated as exhausted for this scope.
expansion.S25 (IL-2)	50	ng/mL	Design choice	–	Classical T-cell expansion cytokine. Included for expansion drive; dose chosen here.

PATH					VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
arm_adjustment adj-it13-03 [BACH2_K0] S24					14.08	ng/mL	Design choice	–	Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'IL-15 dose' decrease. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to increase this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE, same basis as the IL-7 lever. Recorded as a candidate to test, not a recommendation.. Moved from 25.6 to 14.08 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it12. This value came from a bounded search step, not from any publication, so it is a design choice.
arm_adjustment adj-it16-04 [BACH2_K0] S23					25.6	ng/mL	Design choice	–	Reverted to 25.6 ng/mL: the previous revision moved this to 14.08 and the result made it worse. That direction is now treated as exhausted for this scope.

PATH					VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
arm_adjustment adj-it16-05 [BACH2_KO] S18					11.713	ng/mL	Design choice	–	Search step from the search itself, after scoring the previous move (annotation confidence: inference). Lever: 'TCR/activation signal strength' increase. Basis given for the lever: DIRECTION FLIPPED BY THE SEARCH, not by the lever. The lever asked to decrease this parameter and that direction is exhausted for BACH2_KO, so the opposite is being tried. Original basis given for the lever: BACH2 knockout: INFERENCE from roychoudhuri2016: BACH2 opposes AP-1-driven effector genes, so without it an identical activation stimulus delivers a relatively stronger effector push. Lowering it may keep the knockout arm less differentiated. Untested here.. Moved from 7.3206 to 11.713 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it15. This value came from a bounded search step, not from any publication, so it is a design choice.

PATH					VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
arm_adjustment adj-it21-05 [BACH2_K0] S19					25.6	ng/mL	Design choice	–	Reverted to 25.6 ng/mL: the previous revision moved this to 40.96 and the result made it worse. That direction is now treated as exhausted for this scope.
arm_adjustment adj-it21-06 [BACH2_K0] S20					80	ng/mL	Design choice	–	Search step from unimplicated_search. Lever: 'SCF (maturation)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 50 to 80 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it20. This value came from a bounded search step, not from any publication, so it is a design choice.

PATH				VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
arm_adjustment	adj-it22-06	[BACH2_K0]	S12	800	ng/mL	Design choice	–	Search step from unimplicated_search. Lever: 'DLL4 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 500 to 800 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it21. This value came from a bounded search step, not from any publication, so it is a design choice.

PATH	VALUE	UNIT	PROVENANCE	CLAIMS	RATIONALE
arm_adjustment adj-it25-08 [BACH2_K0] S15	16	ng/mL	Design choice	–	Search step from unimplicated_search. Lever: 'IL-7 (t_commitment)' increase. Basis given for the lever: NOT IMPLICATED BY ANY EVIDENCE OR ANNOTATION. The levers the analysis and the bioinformatics tools named are exhausted, so the search is exploring a parameter nothing pointed it at. Treat a gain here as a search result, not as a finding about this parameter.. Moved from 10 to 16 ng/mL for BACH2_KO only. Brief: brief-ipsc-tcell-bach2-d40-001-it24. This value came from a bounded search step, not from any publication, so it is a design choice.

References

Publications

1. Fine-tuning BACH2 dosage balances stemness and effector function to enhance antitumor T cell therapy. Nat Immunol. 2025. [10.1038/s41590-025-02389-z](https://doi.org/10.1038/s41590-025-02389-z)

full text not retrieved

Used for:

- claim L05 (bach2_dosage_effect)

2. Michaels YS, Edgar JM, Major MC, et al. DLL4 and VCAM1 enhance the emergence of T cell-competent hematopoietic progenitors from human pluripotent stem cells. Sci Adv. 2022;8(34):eabn5522.

[10.1126/sciadv.abn5522](https://doi.org/10.1126/sciadv.abn5522) · PMC PMC9401626

full text not retrieved

Used for:

- claim L01 (notch_ligand_requirement)
 - claim L02 (stage_order)
3. Roychoudhuri R, Clever D, Li P, et al. BACH2 regulates CD8+ T cell differentiation by controlling access of AP-1 factors to enhancers. Nat Immunol. 2016;17(7):851-860. [10.1038/ni.3441](https://doi.org/10.1038/ni.3441) · PMC PMC4918801
full text not retrieved

Used for:

- claim L03 (bach2_terminal_differentiation)
4. Tsukumo S, Unno M, Muto A, et al. Bach2 maintains T cells in a naive state by suppressing effector memory-related genes. Proc Natl Acad Sci USA. 2013;110(26):10735-10740. [10.1073/pnas.1306691110](https://doi.org/10.1073/pnas.1306691110) · PMC PMC3696756
full text not retrieved

Used for:

- claim L04 (bach2_naive_maintenance)

Annotation sources

Each annotation carries its own confidence. `local_annotation` restates what a cited paper reports; `inference` is a transfer to this cell type or readout that no cited paper measures; `synthetic_fixture` is invented for software testing.

1. Roychoudhuri et al. 2013, Nature 498:506-510, doi:10.1038/nature12199 (PMID 23728300) - reference key roychoudhuri2013 [tcell_curated_genes]
 - BACH2 knockout: increase target_marker_pct [local_annotation]
2. Roychoudhuri et al. 2016, Nat Immunol 17:851-860, doi:10.1038/ni.3441 (PMID 27158840) - reference key roychoudhuri2016 [tcell_curated_genes]
 - BACH2 knockout: increase growth_rate_per_h [local_annotation]
3. Transfer by the curator from roychoudhuri2016 (doi:10.1038/ni.3441) and bach2_dosage_2025 (doi:10.1038/s41590-025-02389-z) to an iPSC-derived expansion readout. No cited paper measures this. [tcell_curated_genes]
 - BACH2 knockout: decrease target_cells_per_input_cell [inference]
4. Tsukumo et al. 2013, PNAS 110:10735-10740, doi:10.1073/pnas.1306691110 (PMID 23754397) - reference key tsukumo2013 [tcell_curated_genes]
 - BACH2 knockout: decrease target_marker_pct [local_annotation]

Limitations

-
- The reactor was a synthetic stand-in, so no number in this report is a measurement of any real cell population. The stand-in is phenomenological and is not a validated digital twin.
 - Every decision here was authored by the deterministic policy. The reasoning shown is a description of a rule in the code, not a model's reasoning. No LLM ran in this loop.
 - Not one quantity in the final protocol is attributed to a publication: all 36 of them are design choices. The citations support the direction of a lever and nothing numeric.

- 1 replicate per arm. Every between-arm difference is directional only, and the search was climbing a surface it could not distinguish from noise below its own step size.
- A loop that reaches a target has not produced a validated or released process. Confirmation runs, a replicate design and human QA sign-off are all outside it.
- A designed schedule is not an optimised or validated schedule.
- Designed, not extracted. No quantity is attributed to a publication.
- Revised by the deterministic coordinate search in biosense.production.revise. Every revised quantity is a design choice, and no searched value is attributed to a publication.
- Search finding: IL-7 cannot be set to one shared value: moving it 16 -> 25.6 ng/mL improved BACH2_KO and degraded WT. The arms are being given separate values of this parameter from here on.

Generated by biosense.production.report from the JSON in example-ipsc-tcell-bach2. Nothing in this report was written by hand, and nothing in it is clinical or manufacturing guidance.